

Electricity

① Quantization of Charges

Charge $Q = ne$ → Charge on $e^- (1.6 \times 10^{-19} \text{C})$

↓ Coulombs ↓ no. of electrons

② Potential Difference / Voltage (V)

$$V = \frac{W}{Q} \rightarrow \frac{J}{C}$$

J/C or Volt (SI)

→ Voltmeter & i) Parallel
— (V) — ii) High Resistance

③ Electric Current (I) :-

$$I = \frac{Q}{t} \rightarrow \frac{C}{s}$$

→ C/s or Ampere (A) (SI)

Ammeter & i) Series ii) Low Resistance
— (A) —

→ $1 \text{ mA} = 10^{-3} \text{ A}$

→ $1 \mu\text{A} = 10^{-6} \text{ A}$

↑ milli ↓ micro

④ Ohm's Law &

→ $V \propto I$ (At const. temp)

Result $V = IR$

↓ Ohm (Ω)

⑤ Factors affecting resistance

i) $R \propto l$ ii) $R \propto \frac{1}{A}$

iii) $R \propto T$ iv) material of conductor.

⑤ Resistivity (ρ) :-

$$\rho = \frac{R \times A}{l}$$

↓ ($\Omega\text{-m}$)

→ Resistivity depends on material.

⑥ Heating Effect :-

$$H = I^2 \times R \times t$$

Unit - Joule (J).

→ $W = Q \times V$

⑦ Power & (P) SI unit = watt (W)

i) $P = VI$

ii) $P = \frac{V^2}{R}$

iii) $P = I^2 R$

⑧ Electric Energy (E) & KWH

$$E = P \times t$$

1 Kwh = 1 unit

1 Kwh = 3.6×10^6 Joule

⑨ $R_p = \frac{R}{n}$ → $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$ (in Resistance all value)

→ no. of resistance

Total resistance in parallel.

⑩ Fuse &

→ safety device

→ Series

→ M.P. Low

→ Resistance high